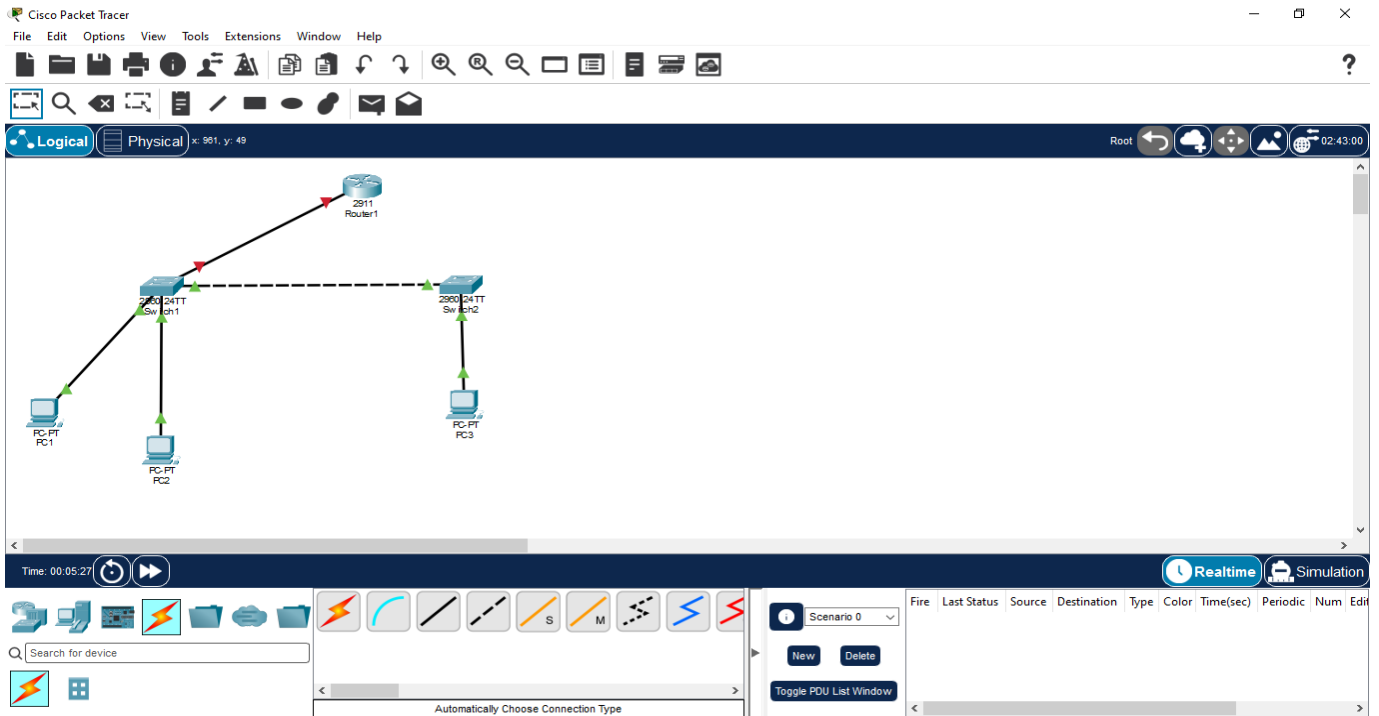


Week 1

Name Makhura Kamogelo

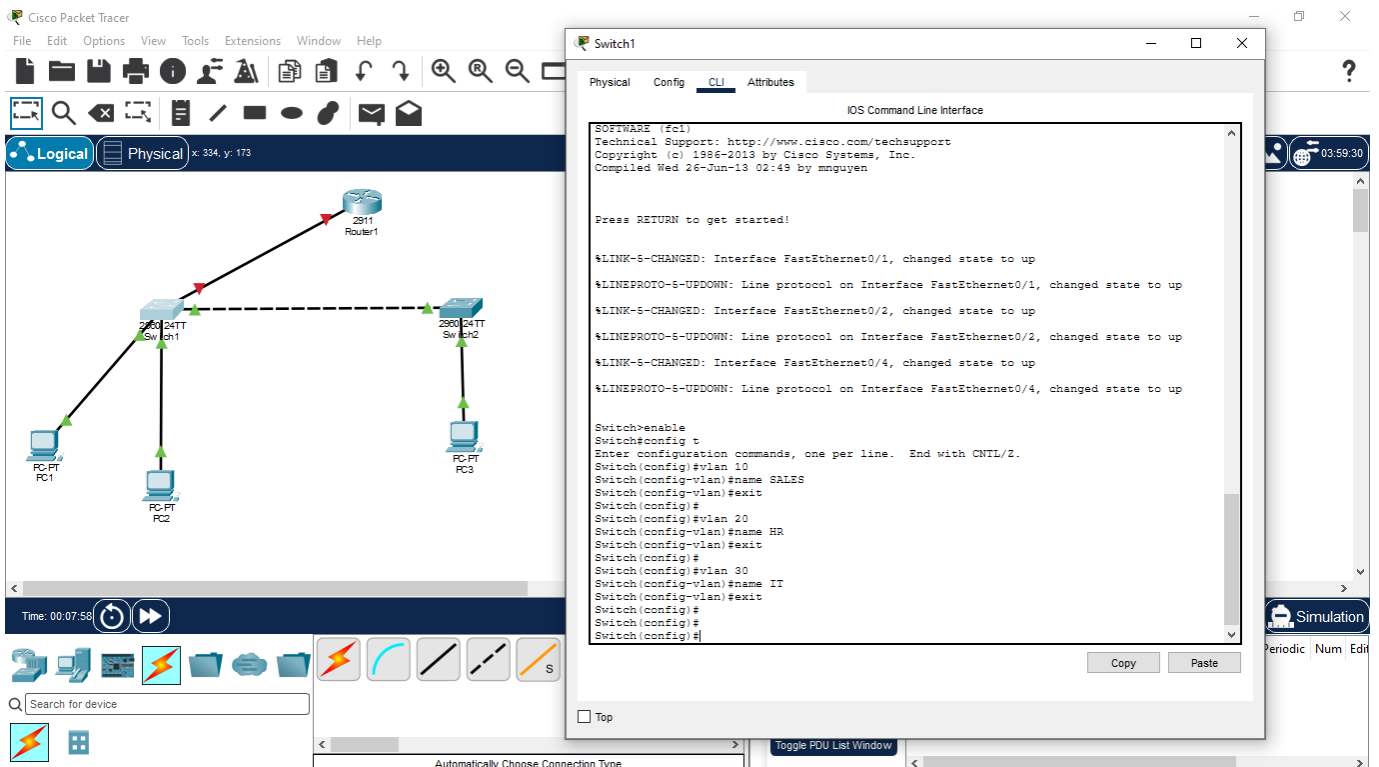
Week 1 practical



Tasks:

Step 1: Create VLANs on SW1

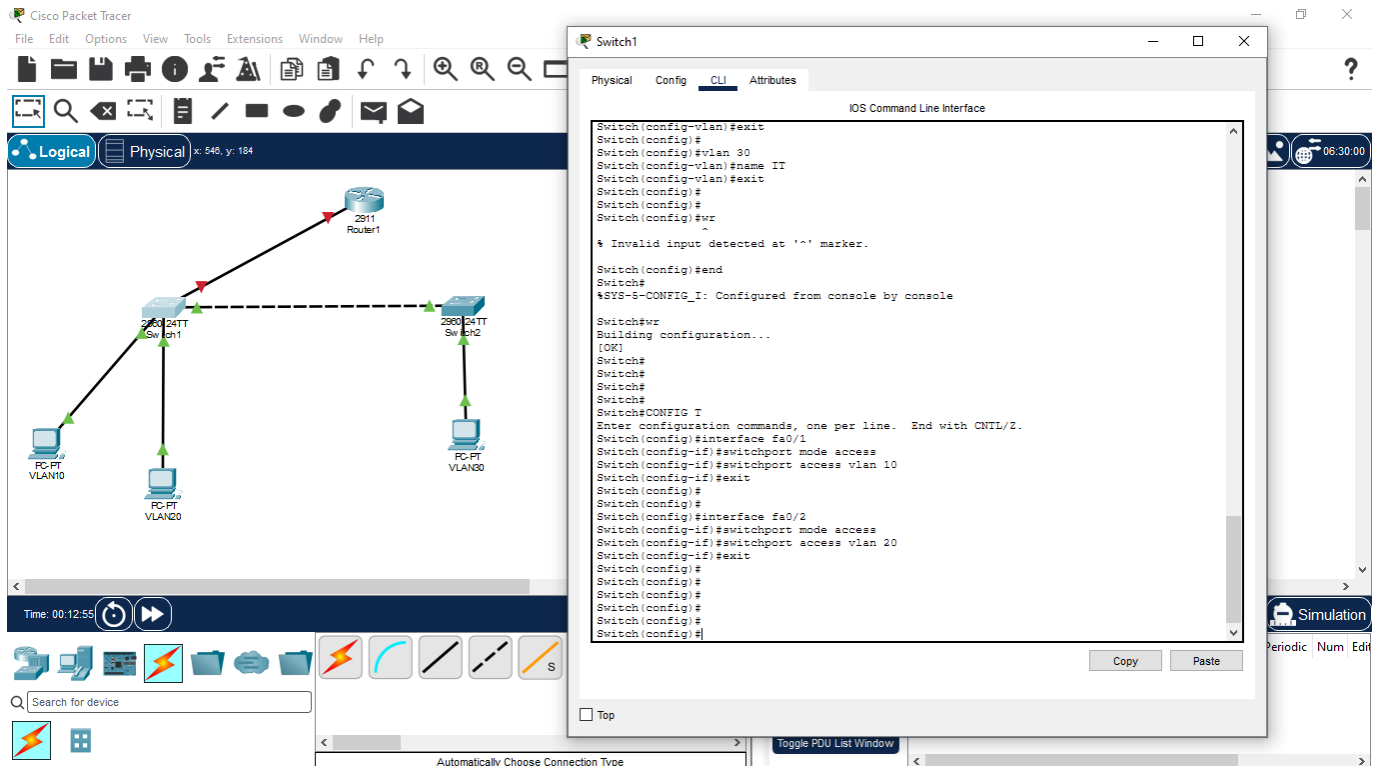
1. vlan 10 name SALES exit vlan 20 name HR exit vlan 30 name IT exit



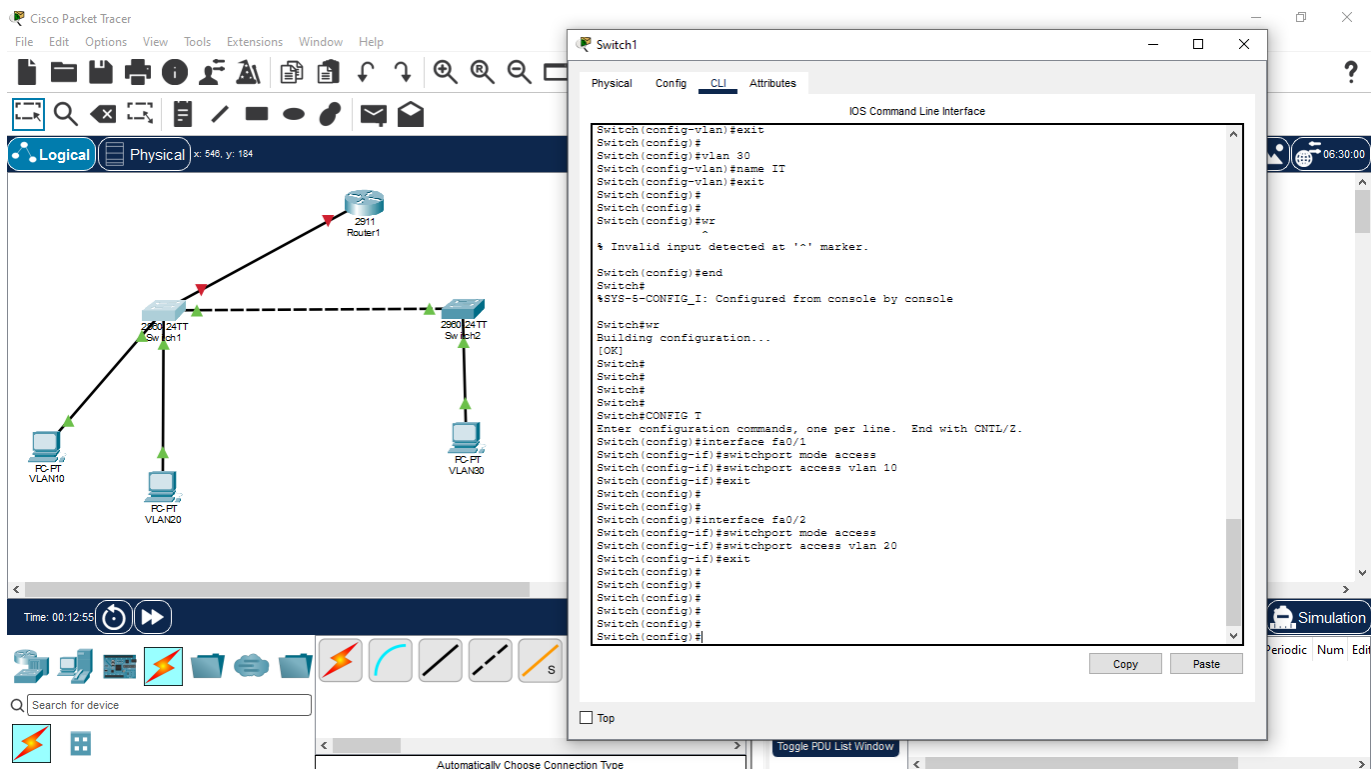
Week 1

Step 2: Assign Access Ports to VLANs

1. interface fa0/1 switchport mode access switchport access vlan 10 exit

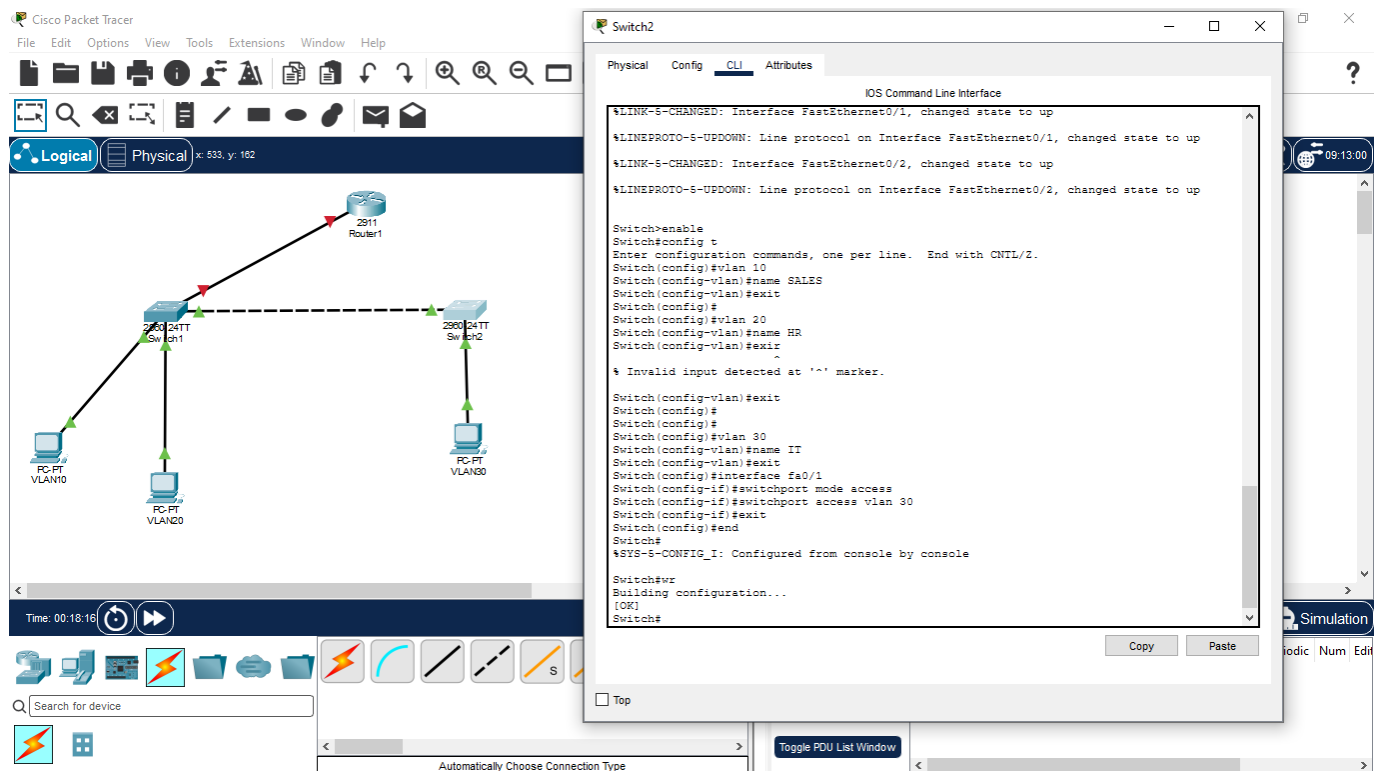


2. interface fa0/2 switchport mode access switchport access vlan 20 exit

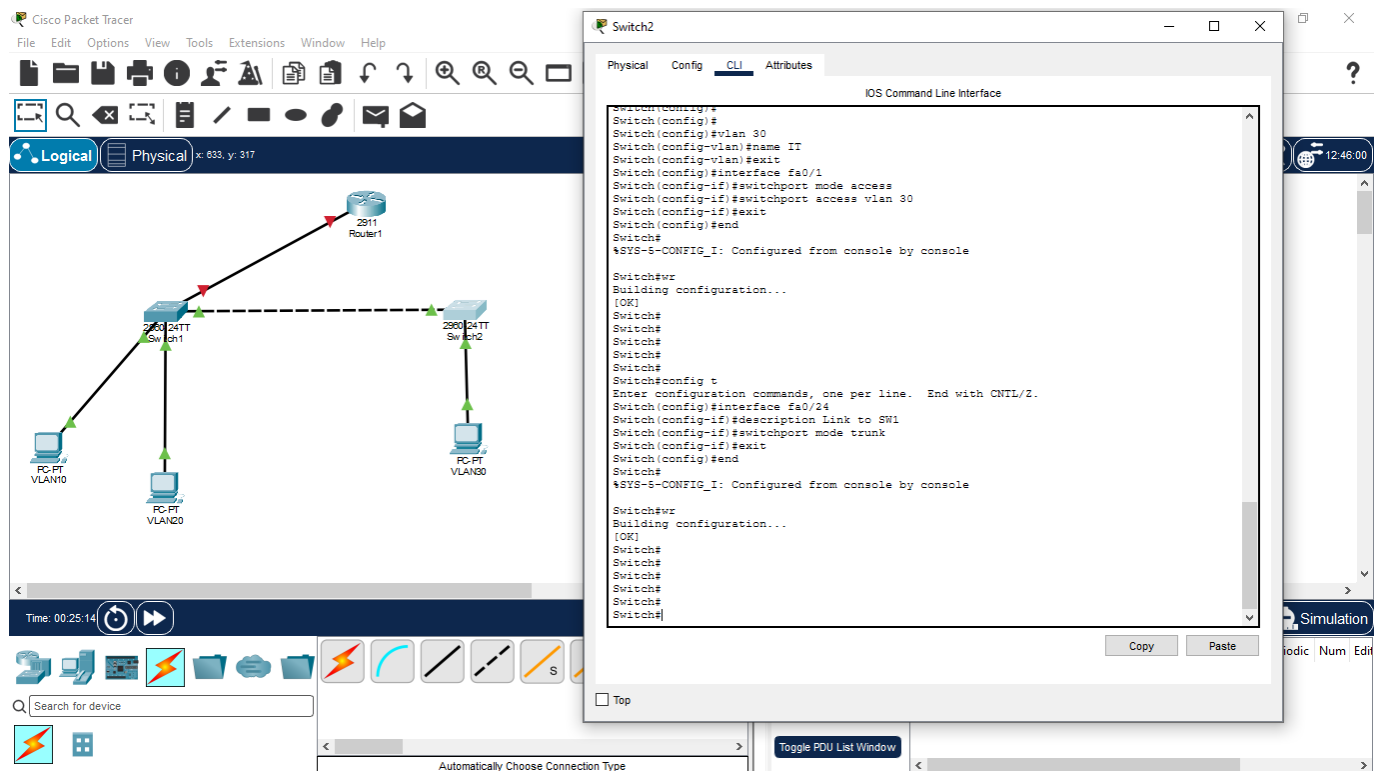


Week 1

3. interface fa0/3 switchport mode access, switchport access vlan 30 exit



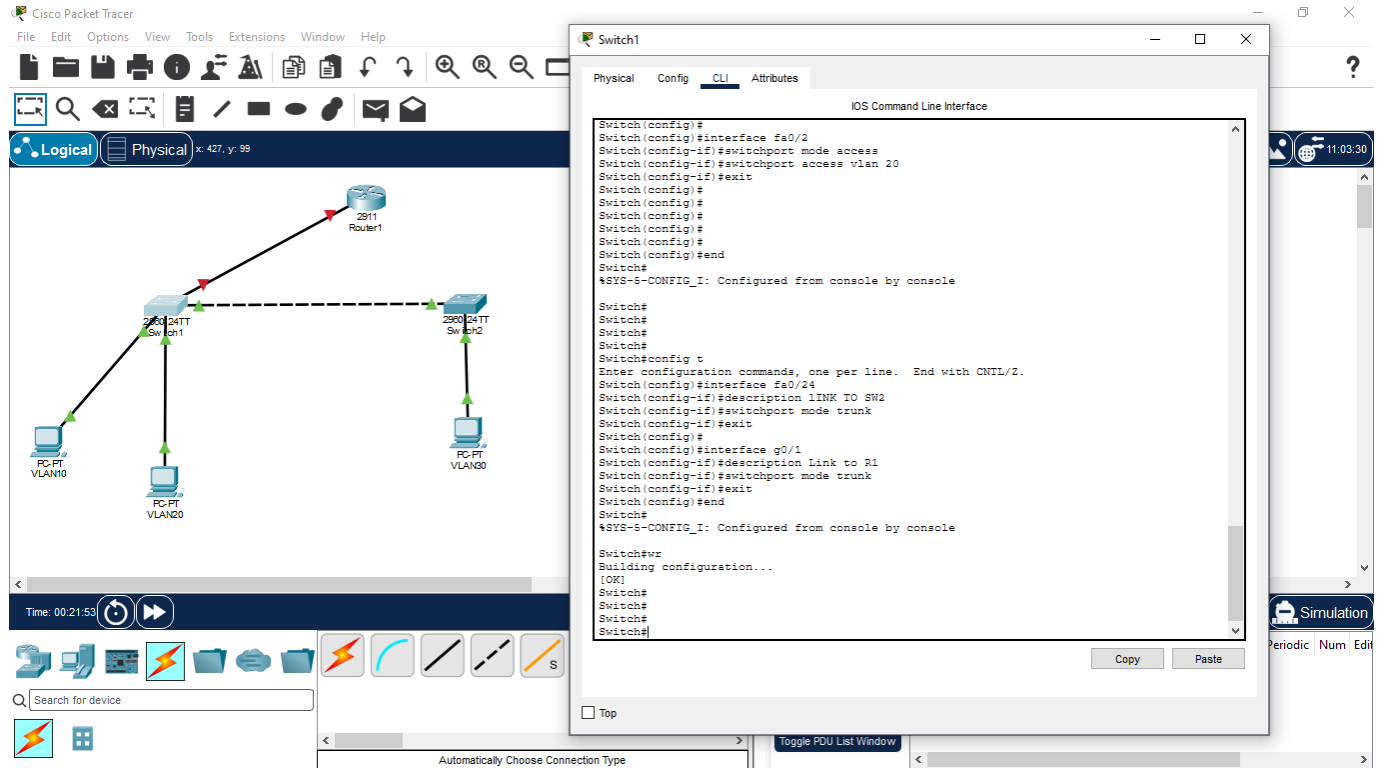
- 4.Repeat VLAN creation and any needed access-port assignments on SW2.



Week 1

Step 3: Configure Trunk Links

1. On the SW1–SW2 link and the SW1–R1 link: interface fa0/24 switchport mode trunk exit

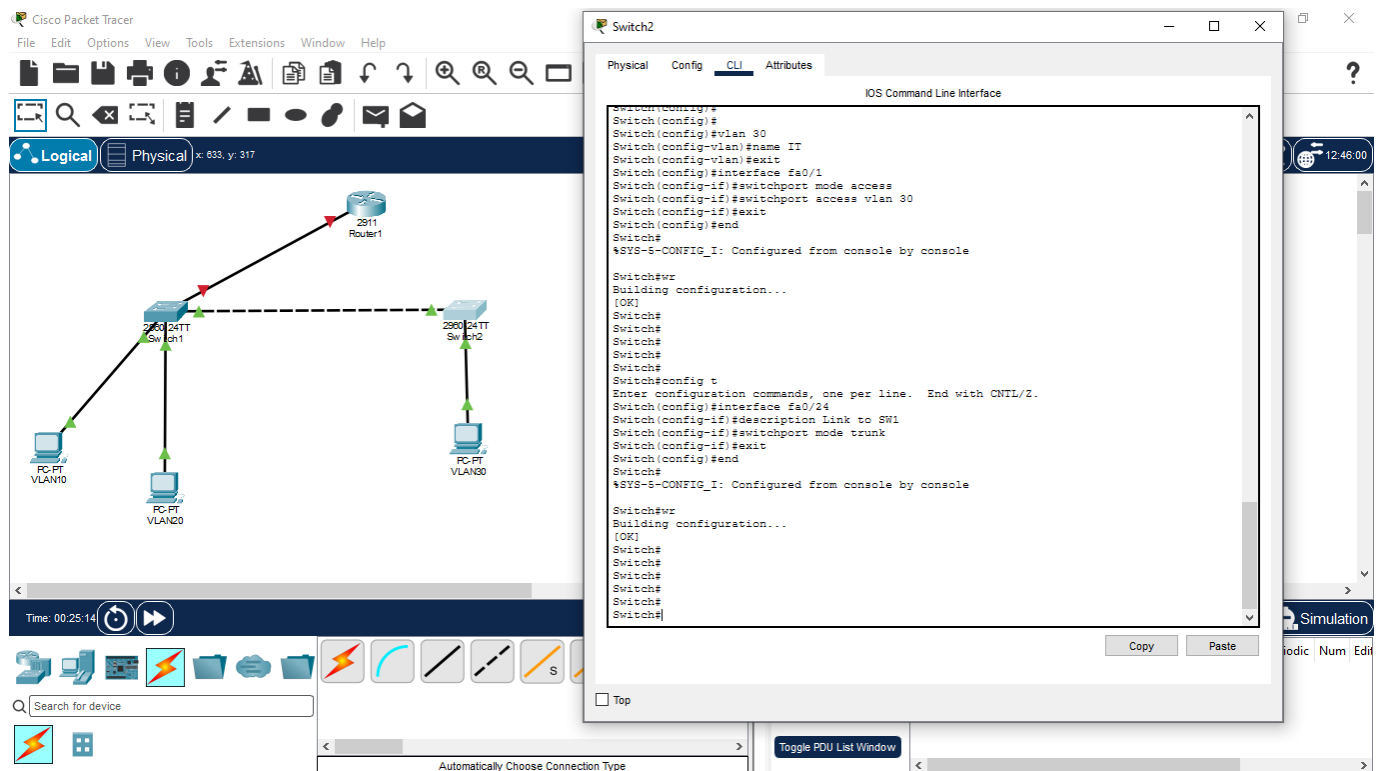


The screenshot shows the Cisco Packet Tracer interface with the CLI window for Switch1 open. The network diagram on the left shows a topology with a 2911 Router connected to Switch1, which is connected to Switch2. Switch1 is also connected to two PCs (VLAN10 and VLAN20). The CLI window for Switch1 shows the configuration process for interface fa0/24, setting it to access mode and then trunk mode.

```
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#
Switch#
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#description LINK TO SW2
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface g0/1
Switch(config-if)#description Link to R1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
Switch#
Switch#
Switch#
```



The screenshot shows the Cisco Packet Tracer interface with the CLI window for Switch2 open. The network diagram on the left is the same as the previous screenshot. The CLI window for Switch2 shows the configuration process for interface fa0/24, setting it to access mode and then trunk mode.

```
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#name II
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

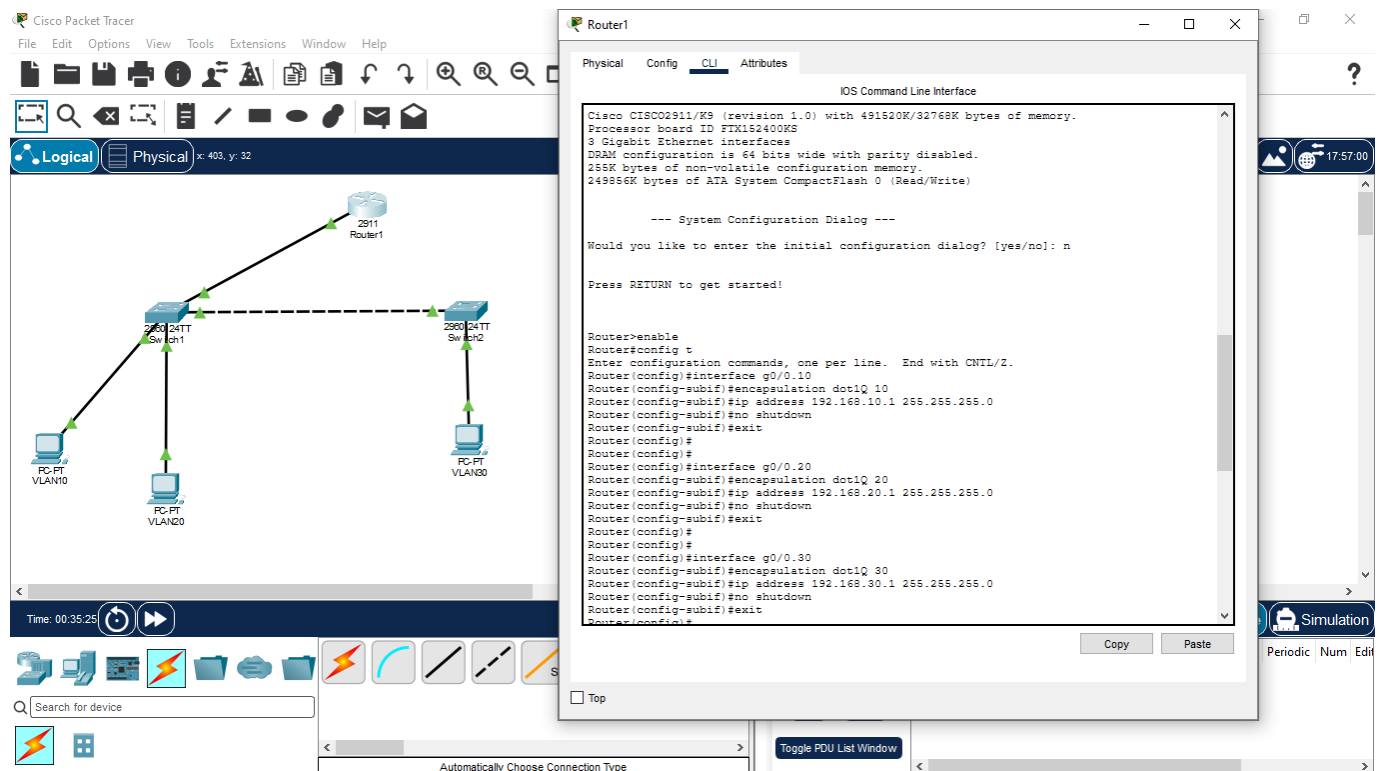
Switch#wr
Building configuration...
[OK]
Switch#
Switch#
Switch#
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#description Link to SW1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
Switch#
Switch#
Switch#
Switch#
```

Week 1

Step 4: Configure Router-on-a-Stick on R1

1. interface g0/0.10 encapsulation dot1Q 10 ip address 192.168.10.1 255.255.255.0
2. interface g0/0.20 encapsulation dot1Q 20 ip address 192.168.20.1 255.255.255.0
3. interface g0/0.30 encapsulation dot1Q 30 ip address 192.168.30.1 255.255.255.0



Step 5: Verify

1. show vlan brief
2. show interfaces trunk
3. Ping between two PCs in different VLANs to confirm R1 is routing between them.

Week 1

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 613, y: 161

Time: 00:40:22

Search for device

Automatically Choose Connection Type

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>enable
Switch#show vlan brief

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 SALES	active	Fa0/1
20 HR	active	Fa0/2
30 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Copy Paste

Toggle PDU List Window

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 621, y: 191

Time: 00:41:03

Search for device

Automatically Choose Connection Type

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

Switch>enable
Switch#show vlan brief

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 SALES	active	
20 HR	active	
30 IT	active	Fa0/1
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Copy Paste

Toggle PDU List Window

Week 1

Reflection Questions (answer these in your submission):

1. Why can't PC-SALES (VLAN 10) and PC-HR (VLAN 20) ping each other without R1, even though they may be plugged into the same switch?

Because VLANs create separate Layer 2 broadcast domains.

A switch only forwards traffic within the same VLAN. PC-SALES in VLAN10 and PC-HR in VLAN20 are logically on 2 different networks, even if physically on SW1.

The switch does not route between VLANs-it only switches. To communicate between different VLANs and different IP subnets, traffic must go to a Layer 3 device like router R1. R1 strips the VLAN tag, routes the packet, and puts it back into the destination VLAN.